The Science of Salsa: Antimicrobial Properties of Salsa Components to Learn Scientific Methodology

Most ethnic foods and cooking practices have incorporated the use of spices and other food additives. Many common spices have crossed cultural boundaries and appear in multiple ethnic cuisines. Recent studies have demonstrated that many of these ingredients possess antimicrobial properties against common food spoilage microorganisms. We developed a laboratory exercise that promotes the use of scientific methodology to evaluate the effectiveness of salsa components at inhibiting the growth of undesirable microorganisms. Tomato, onion, garlic, cilantro, and jalapeño were tested for antimicrobial properties against a representative fungus, Saccharomyces cerevisiae, and the common food spoilage bacteria Staphylococcus aureus, Bacillus cereus, and Escherichia coli. Each component was ethanol extracted and a modification of the Kirby-Bauer method of antimicrobial sensitivity was employed. Garlic demonstrated the greatest inhibitory effects against all organisms tested. Onion demonstrated a slight inhibition of all four organisms, while cilantro showed some inhibition of all three bacteria but no effect against the fungus. Jalapeño may have slightly inhibited E. coli and S. aureus, as evidenced by a consistently measured increase in the zone of inhibition that was not statistically significant when compared to that of the control. Following the initial exercise, students were given the opportunity to repeat the exercise using other spices such as cinnamon, clove, nutmeg, and coriander. Student learning outcomes were evaluated using preliminary and secondary surveys, mainly focusing on definitions of science and hypothesis as well as the process of science. Students enjoyed this exercise and met the learning goals of understanding the process and methodology of science, as well as the interdisciplinarity inherent in the sciences. Student learning was evidenced by an increase in the number of correct responses on the secondary survey in comparison to the preliminary.